

EE 316

Section 3

Pre Lab 01

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Inverting and Non-Inverting Op Amps

Introduction:

This laboratory investigates inverting and non-inverting operational amplifier (op-amp) configurations using both DC and AC inputs. The lab provides experience analyzing and simulating op-amp circuits while examining closed-loop voltage gain and the effects of load resistance on output voltage.

Simulations and Results:

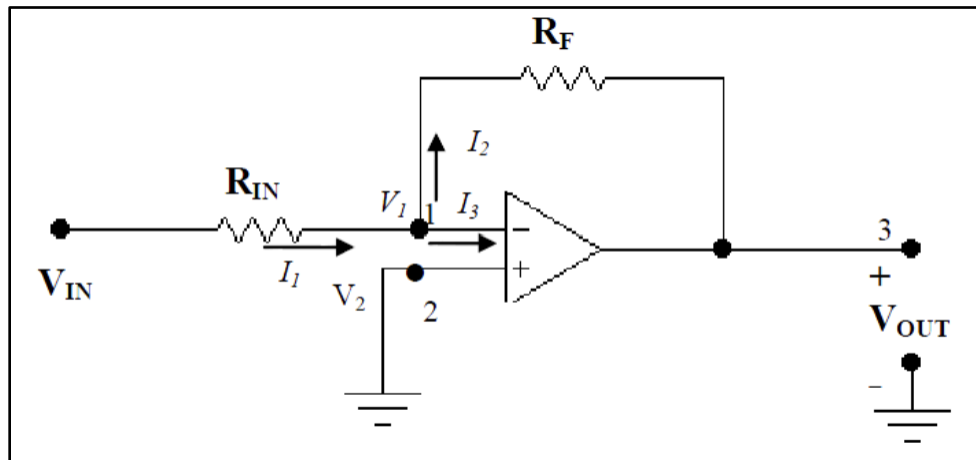


Figure #1: Inverting Op-Amp Schematic from Lab Manual

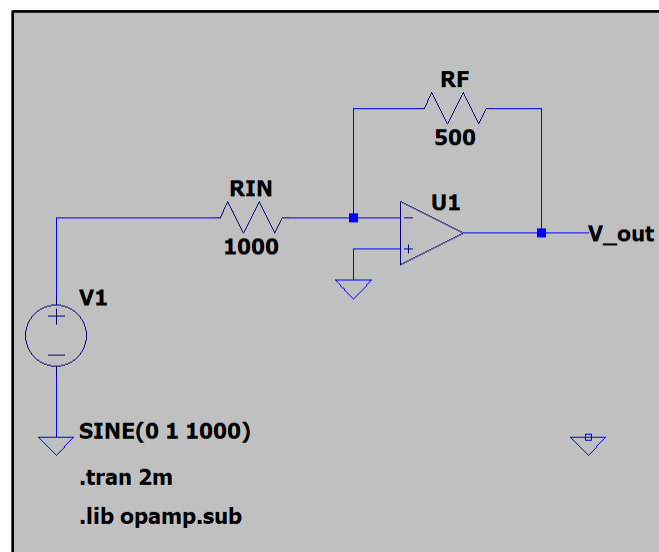


Figure #2: Inverting Op-Amp with $R_F = 500$ Ohms (LTSpice)

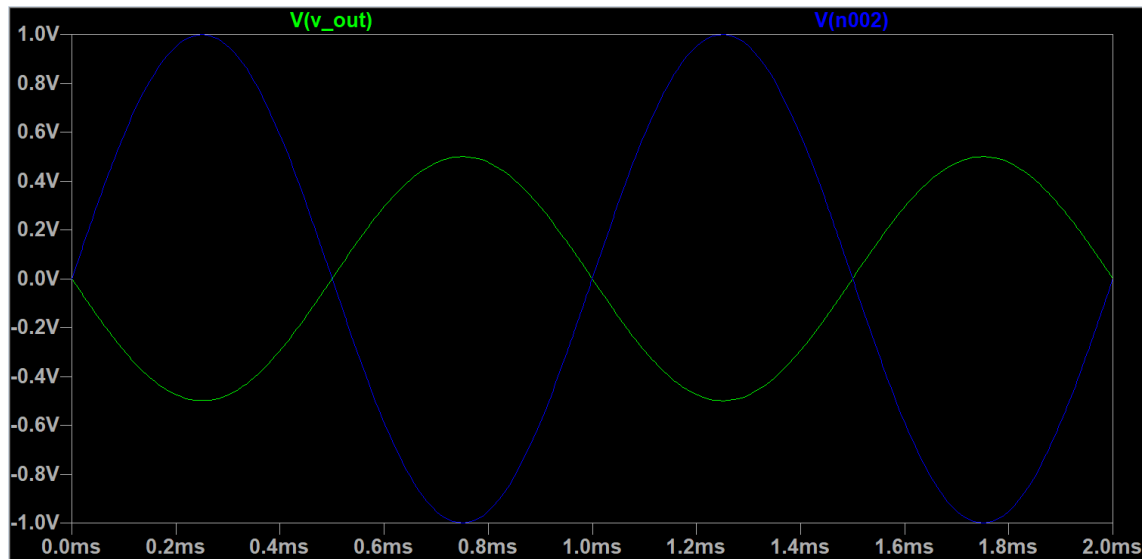


Figure #3: Inverting Op-Amp Oscilloscope Output at $R_F = 500$ Ohms (V_{out} and V_{in})

V_{inpp} (V)	R_{IN} (k Ω)	R_F (k Ω)	V_{outpp} (V) O-scope	Gain (V/V)	V_{outpp} (V) DMM
2	1	0.5	1	-0.5	0.354
		1	2	-1	0.707
		2	4	-2	1.414
		3	6	-3	2.122
		4	8	-4	2.829

Table #1: Inverting Amplifier Simulation Results

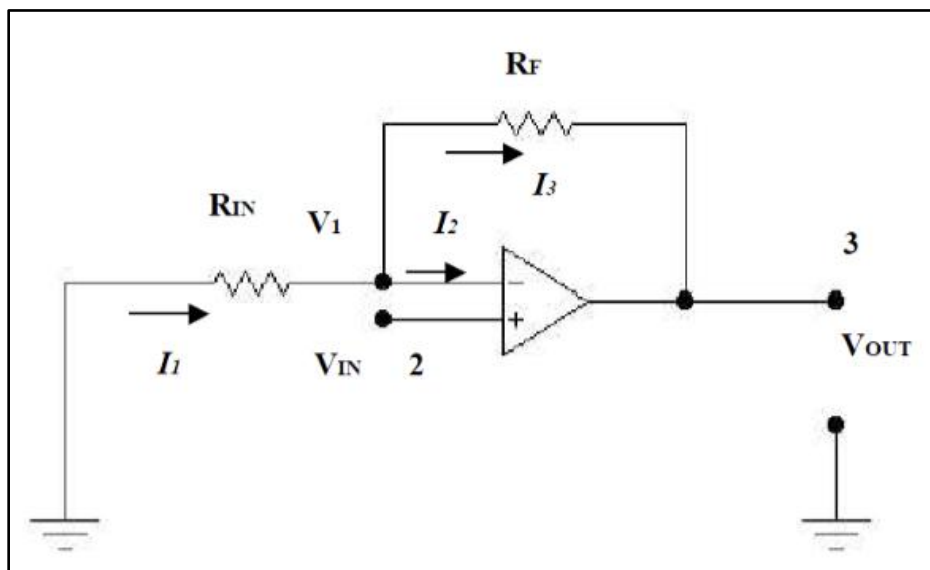


Figure #4: Non- Inverting Op-Amp Schematic from Lab Manual

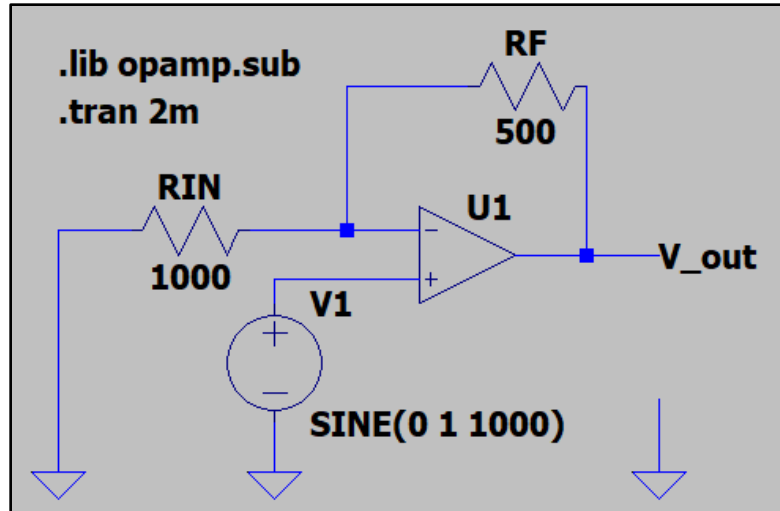


Figure #5: Non-Inverting Op-Amp with $R_F = 500$ Ohms (LTSpice)

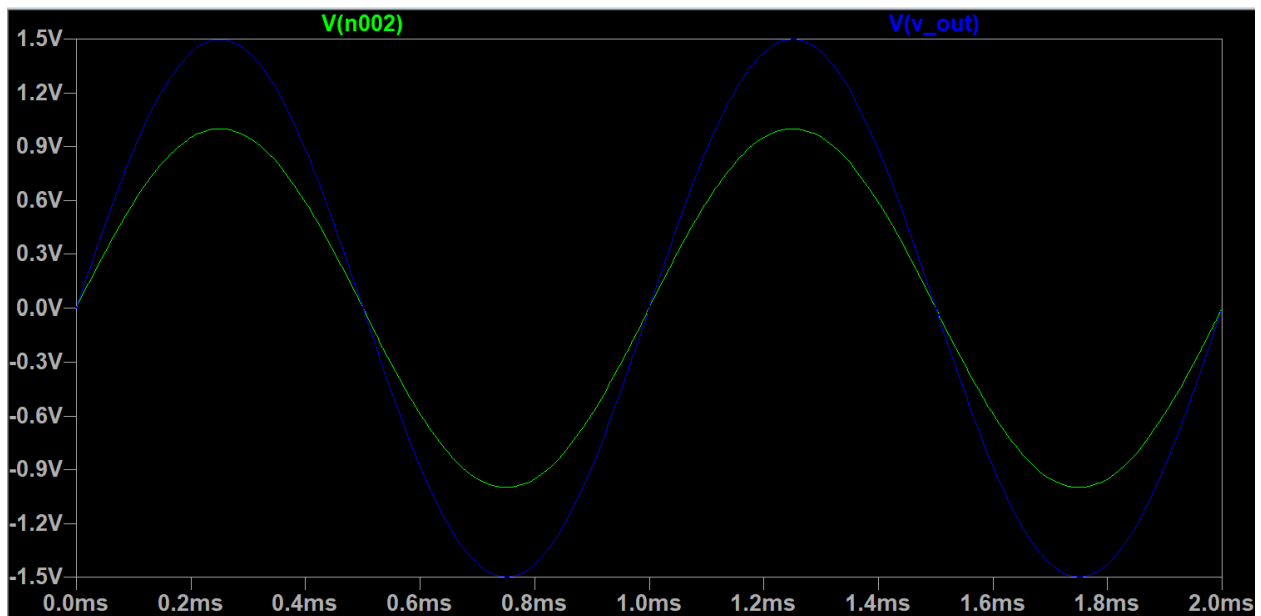


Figure #6: Non-Inverting Op-Amp Oscilloscope Output at $R_F = 500$ Ohms (V_{out} and V_{in})

V_{inpp} (V)	R_{IN} (k Ω)	R_F (k Ω)	V_{outpp} (V) O-scope	Gain (V/V)	V_{outpp} (V) DMM
2	1	0.5	3	1.5	2.121
		1	4	2	2.828
		2	6	3	4.242
		3	8	4	5.657
		4	10	5	7.071

Table #2: Non-Inverting Amplifier Simulation Results